Adaptive control of multi-joint robotic manipulator systems with RBF neural networks

Yiheng Lyu

***Abstract*—This paper proposes an adaptive RBF neural control scheme for trajectory tracking of uncertain robotic manipulators. Unknown nonlinear dynamics are approximated online using radial basis function networks within a computed torque control framework. A σ-modified adaptive law is introduced to improve robustness against parameter drift. Lyapunov analysis and simulation results demonstrate the effectiveness of the proposed method.**

***Index Terms*—Robotic manipulators; adaptive control; RBF neural networks; trajectory tracking; computed torque control.**

I. Introduction

ROBOTIC manipulators have become indispensable in modern manufacturing, industrial automation, medical robotics, and human–robot interaction due to their high precision, flexibility, and autonomous operation capabilities. In many robotic applications, accurate trajectory tracking is essential for ensuring motion precision, operational safety, and task efficiency. However, achieving high-performance tracking remains challenging because practical robotic systems are inevitably affected by modeling uncertainties, external disturbances, and unmodeled nonlinear dynamics. These uncertainties may significantly degrade tracking accuracy and even compromise the stability of conventional model-based control methods.

To improve tracking performance under uncertainties, a variety of nonlinear control approaches have been extensively investigated, including computed torque control, robust control, adaptive control, sliding mode control, and intelligent control methods. Among them, computed torque control has been widely adopted because of its systematic design methodology and excellent tracking performance when accurate dynamic models are available. Nevertheless, its effectiveness strongly depends on precise knowledge of the manipulator dynamics, which is often difficult to obtain in practice due to uncertain physical parameters, friction, payload variations, and external disturbances. Adaptive control alleviates part of this limitation by estimating unknown parameters online, while robust control improves disturbance rejection at the expense of increased control effort or conservative gain selection.

In recent years, neural-network-based adaptive control has attracted considerable attention because of its capability to approximate unknown nonlinear functions without requiring explicit mathematical models. In particular, radial basis function (RBF) neural networks have been widely employed owing to their universal approximation capability, simple network structure, and efficient online learning properties. By integrating RBF neural networks with adaptive control, unknown dynamics can be estimated online and compensated during controller execution, thereby improving tracking accuracy under uncertain environments. Nevertheless, maintaining parameter boundedness and ensuring stable online adaptation remain important issues in adaptive neural control, especially in the presence of approximation errors and persistent disturbances.

Motivated by these observations, this paper investigates the trajectory tracking problem for uncertain robotic manipulators using an adaptive neural control framework based on RBF networks. Unknown nonlinear dynamics are represented as lumped uncertainties and approximated online through RBF neural networks within a computed torque control architecture. Furthermore, a σ-modified adaptive law is incorporated to suppress parameter drift and enhance robustness during online learning. Stability of the overall control system is established through Lyapunov analysis, and numerical simulations are conducted to evaluate the effectiveness of the proposed approach.

The main contributions of this paper are summarized as follows.

An adaptive neural control framework combining computed torque control and RBF neural network approximation is developed for trajectory tracking of uncertain robotic manipulators. A σ-modified adaptive law is incorporated into the online learning process to improve parameter robustness and suppress weight drift. Lyapunov-based stability analysis and numerical simulations demonstrate the effectiveness of the proposed control strategy under model uncertainties and external disturbances.

The remainder of this paper is organized as follows. Section II introduces the preliminary knowledge of robotic manipulator dynamics and RBF neural networks. Section III formulates the trajectory tracking problem. Section IV presents the adaptive controller design together with the corresponding stability analysis. Section V provides numerical simulation results. Finally, Section VI concludes the paper.

II. Preliminaries

A. Radial Basis Function (RBF) Neural Networks

In this paper, Radial Basis Function (RBF) neural networks (NNs) are utilized to approximate the unknown continuous non-linear functions within the robotic manipulator dynamics.

Generally, an RBF NN can be represented as the linear combination of localized radial basis functions. Given an input

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TABLE I: Dimensions of the Main Variables

Variable	Dimension
$e, \dot{e}$	$\mathbb{R}^{n\times 1}$
$X = \begin{bmatrix} e \\ \dot{e} \end{bmatrix}$	$\mathbb{R}^{2n\times 1}$
P	$\mathbb{R}^{2n\times 2n}$
$B = \begin{bmatrix} 0 \\ M_0^{-1}(q) \end{bmatrix}$	$\mathbb{R}^{2n\times n}$
ε	$\mathbb{R}^{n\times 1}$
$h(X)$	$\mathbb{R}^{m\times 1}$
$\hat{W}$	$\mathbb{R}^{m\times n}$

vector $x \in \mathbb{R}^m$, the actual output of the RBF NN is expressed in a linearly parameterized form as:

$$\hat{f}(x) = \hat{W}^T h(x) \tag{1}$$

where $\hat{W} \in \mathbb{R}^{m\times p}$ is the weight matrix, and $h(x) = [h_1(x), h_2(x), \ldots, h_m(x)]^T \in \mathbb{R}^m$ is the hidden layer activation vector. The Gaussian function is adopted as the basis function for the i-th hidden node:

$$h_i(x) = \exp\left(-\frac{\|x - c_i\|^2}{2b_i^2}\right), \quad i = 1, 2, \ldots, n \tag{2}$$

where $c_i \in \mathbb{R}^m$ and $b_i > 0$ denote the center vector and the width parameter, respectively.

According to the universal approximation theorem, for any continuous function $f(x)$ defined on a compact set $\Omega_x \subset \mathbb{R}^m$, there exists an ideal constant weight matrix W^* such that:

$$f(x) = {W^*}^T h(x) + \epsilon \tag{3}$$

where $\epsilon \in \mathbb{R}^p$ represents the inherent NN approximation error. The ideal weight matrix W^* is defined as:

$$W^* \triangleq \arg\min_{W} \left[\sup_{x\in\Omega_x} \|f(x) - W^T h(x)\|\right] \tag{4}$$

Furthermore, the approximation error ϵ is assumed to be bounded on the compact set Ω_x, satisfying:

$$\|\epsilon\| \leq \epsilon_N \tag{5}$$

where $\epsilon_N > 0$ is an unknown positive constant.

Remark 1: The linearly parameterized structure of the RBF NN implies that the network output is linear with respect to the weight parameter $\hat{W}$, while maintaining a non-linear mapping via $h(x)$. This characteristic significantly facilitates the derivation of adaptive parameter update laws using the Lyapunov synthesis method, thereby ensuring the closed-loop stability of the robotic system.

B. Notation

The dimensions of the principal variables are summarized in Table I.

III. Problem Formulation

A. System Dynamics

Consider an n-degree-of-freedom (DOF) rigid robotic manipulator governed by the Euler–Lagrange equation

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = \tau + d, \tag{6}$$

where $q, \dot{q}, \ddot{q} \in \mathbb{R}^n$ denote the joint position, velocity, and acceleration vectors, respectively. The matrix $M(q) \in \mathbb{R}^{n\times n}$ denotes the inertia matrix, $C(q,\dot{q}) \in \mathbb{R}^{n\times n}$ is the Coriolis and centrifugal matrix, $G(q) \in \mathbb{R}^n$ is the gravity vector, $\tau \in \mathbb{R}^n$ is the control input, and $d \in \mathbb{R}^n$ represents the lumped external disturbance and unmodeled dynamics.

The following standard properties of Euler–Lagrange systems are adopted throughout this paper.

Property 1: The inertia matrix $M(q)$ is symmetric and uniformly positive definite. Specifically, there exist two positive constants m_1 and m_2 satisfying

$$m_1 I \leq M(q) \leq m_2 I, \tag{7}$$

for all $q \in \mathbb{R}^n$.

Property 2: The matrix $\dot{M}(q) - 2C(q,\dot{q})$ is skew-symmetric, i.e.,

$$x^{\mathrm{T}}\left(\dot{M}(q) - 2C(q,\dot{q})\right)x = 0, \tag{8}$$

for any $x \in \mathbb{R}^n$.

In practical robotic systems, the exact dynamic model is unavailable due to parameter uncertainties, payload variations, friction, and unmodeled dynamics. Accordingly, the manipulator dynamics are decomposed into a nominal model and an uncertainty term as

$$M(q) = M_0(q) + \Delta M(q), \tag{9}$$

$$C(q,\dot{q}) = C_0(q,\dot{q}) + \Delta C(q,\dot{q}), \tag{10}$$

$$G(q) = G_0(q) + \Delta G(q), \tag{11}$$

where $M_0(q)$, $C_0(q,\dot{q})$, and $G_0(q)$ denote the nominal inertia, Coriolis/centrifugal, and gravity terms, respectively.

Substituting the above decompositions into (6) yields

$$(M_0 + \Delta M)\,\ddot{q} + (C_0 + \Delta C)\,\dot{q} + (G_0 + \Delta G) = \tau + d. \tag{12}$$

Rearranging the above equation gives

$$M_0(q)\ddot{q} + C_0(q,\dot{q})\dot{q} + G_0(q) = \tau + f(q,\dot{q},\ddot{q}), \tag{13}$$

where

$$f(q,\dot{q},\ddot{q}) = d - \Delta M(q)\ddot{q} - \Delta C(q,\dot{q})\dot{q} - \Delta G(q). \tag{14}$$

The nonlinear function $f(\cdot)$ represents the lumped uncertainty consisting of modeling errors, parameter variations, external disturbances, and unmodeled dynamics.

B. Tracking Control Problem

Assume that the desired joint trajectory satisfies

$$q_d,\ \dot{q}_d,\ \ddot{q}_d \in \mathcal{L}_\infty. \tag{15}$$

Define the tracking error and its derivative as

$$e = q - q_d, \qquad \dot{e} = \dot{q} - \dot{q}_d. \tag{16}$$

Differentiating (16) yields

$$\ddot{e} = \ddot{q} - \ddot{q}_d. \tag{17}$$

The desired tracking performance is specified by the second-order error dynamics

$$\ddot{e} + K_v\dot{e} + K_p e = 0, \tag{18}$$

where $K_p, K_v \in \mathbb{R}^{n\times n}$ are symmetric positive-definite gain matrices.

Assumption 1: The external disturbance is bounded, i.e.,

$$\|d(t)\| \le d_{\max}, \tag{19}$$

where $d_{\max}$ is an unknown positive constant.

Substituting (17) into (18) yields the desired joint acceleration

$$\ddot{q} = \ddot{q}_d - K_v\dot{e} - K_p e. \tag{20}$$

The control objective is to design an adaptive neural controller such that all closed-loop signals remain bounded and the tracking error converges to a sufficiently small neighborhood of the origin despite the presence of lumped uncertainties.

IV. Controller Design

A. Nominal Controller Design

To achieve the desired tracking objective, define the auxiliary control input as

$$v = \ddot{q}_d - K_v\dot{e} - K_p e, \tag{21}$$

where $K_p, K_v \in \mathbb{R}^{n\times n}$ are symmetric positive-definite gain matrices.

Based on the nominal manipulator dynamics, the control law is designed as

$$\tau = M_0(q)v + C_0(q,\dot{q})\dot{q} + G_0(q) - \hat{f}(x), \tag{22}$$

where $\hat{f}(x)$ denotes the neural-network estimation of the lumped uncertainty.

Remark 2: The introduction of the virtual control input v decouples the controller into two functional modules. The outer-loop controller computes the virtual acceleration command v to achieve the desired tracking performance, whereas the inner-loop controller realizes this command by generating the actual joint torque τ through dynamic inversion and RBF neural network compensation. This modular design separates trajectory tracking from dynamic compensation, thereby simplifying controller design and implementation.

Substituting (22) into the nominal dynamics (13) yields

$$M_0(q)\ddot{q} = M_0(q)v + f(q,\dot{q},\ddot{q}) - \hat{f}(x). \tag{23}$$

Using the definition of v in (21), one obtains

$$M_0(q)\left(\ddot{q} - \ddot{q}_d + K_v\dot{e} + K_p e\right) = f - \hat{f}(x). \tag{24}$$

Since

$$\ddot{e} = \ddot{q} - \ddot{q}_d, \tag{25}$$

the closed-loop tracking error dynamics can be written as

$$\ddot{e} + K_v\dot{e} + K_p e = M_0^{-1}(q)\left(f - \hat{f}(x)\right). \tag{26}$$

B. State-Space Representation

Define the state vector as

$$X = \begin{bmatrix} e \\ \dot{e} \end{bmatrix}, \qquad \dot{X} = \begin{bmatrix} \dot{e} \\ \ddot{e} \end{bmatrix}. \tag{27}$$

Substituting (26) into (27) gives

$$\dot{X} = AX + B\left(f - \hat{f}(x)\right), \tag{28}$$

where

$$A = \begin{bmatrix} 0 & I \\ -K_p & -K_v \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ M_0^{-1}(q) \end{bmatrix}. \tag{29}$$

Equation (28) reformulates the second-order closed-loop system as a first-order state-space representation, facilitating the subsequent stability analysis and adaptive controller design.

C. RBF Neural Network Approximation

To compensate for the unknown lumped uncertainty $f(x)$, an RBF neural network is employed.

According to the universal approximation theorem, there exists an ideal constant weight matrix W^* such that

$$f(x) = W^{*T}h(x) + \varepsilon, \tag{30}$$

where $h(x) \in \mathbb{R}^m$ denotes the Gaussian basis function vector, $W^* \in \mathbb{R}^{m\times n}$ is the ideal weight matrix, and $\varepsilon \in \mathbb{R}^n$ represents the bounded approximation error satisfying

$$\|\varepsilon\| \le \bar{\varepsilon}. \tag{31}$$

The neural-network output is given by

$$\hat{f}(x) = \hat{W}^T h(x), \tag{32}$$

where $\hat{W}$ denotes the online estimate of the ideal weight matrix.

Define the weight estimation error as

$$\tilde{W} = W^* - \hat{W}. \tag{33}$$

Substituting (30) and (32) into (28) yields

$$\dot{X} = AX + B\left(\tilde{W}^T h(x) + \varepsilon\right), \tag{34}$$

which will be employed in the subsequent Lyapunov stability analysis.

V. Stability Analysis

In this section, the stability of the proposed adaptive neural controller is established using the Lyapunov direct method. For simplicity of the theoretical analysis, the nominal closed-loop system matrix A defined in (29) is assumed to be Hurwitz. Consequently, for any given symmetric positive-definite matrix $Q \in \mathbb{R}^{2n\times 2n}$, there exists a unique symmetric positive-definite matrix $P \in \mathbb{R}^{2n\times 2n}$ satisfying the Lyapunov equation

$$A^T P + PA = -Q, \tag{35}$$

where $Q = Q^T > 0$ and $P = P^T > 0$.

A. Lyapunov Stability Analysis

Consider the following composite Lyapunov candidate function

$$V = \frac{1}{2}X^T PX + \frac{1}{2\gamma}\operatorname{tr}\left(\tilde{W}^T\tilde{W}\right), \tag{36}$$

where $\gamma > 0$ denotes the adaptation gain.

Differentiating (36) with respect to time yields

$$\dot{V} = \frac{1}{2}\dot{X}^T PX + \frac{1}{2}X^T P\dot{X} + \frac{1}{\gamma}\operatorname{tr}\left(\dot{\tilde{W}}^T\tilde{W}\right). \tag{37}$$

Since the ideal weight matrix W^* is constant, one has

$$\dot{\tilde{W}} = -\dot{\hat{W}}. \tag{38}$$

Substituting the closed-loop system (34)

$$\dot{X} = AX + B\left(\tilde{W}^T h(x) + \varepsilon\right) \tag{39}$$

into (37) gives

$$\begin{aligned}\dot{V} =& \frac{1}{2}\left[AX + B\left(\tilde{W}^T h + \varepsilon\right)\right]^T PX \\ &+ \frac{1}{2}X^T P\left[AX + B\left(\tilde{W}^T h + \varepsilon\right)\right] \\ &- \frac{1}{\gamma}\operatorname{tr}\left(\dot{\hat{W}}^T\tilde{W}\right).\end{aligned} \tag{40}$$

Expanding (40) yields

$$\begin{aligned}\dot{V} =& \frac{1}{2}X^T(A^T P + PA)X \\ &+ X^T PB\tilde{W}^T h + X^T PB\varepsilon \\ &- \frac{1}{\gamma}\operatorname{tr}\left(\dot{\hat{W}}^T\tilde{W}\right).\end{aligned} \tag{41}$$

Using the Lyapunov equation (35) $A^T P + PA = -Q$, one obtains

$$\dot{V} = -\frac{1}{2}X^T QX + X^T PB\tilde{W}^T h + X^T PB\varepsilon - \frac{1}{\gamma}\operatorname{tr}\left(\dot{\hat{W}}^T\tilde{W}\right). \tag{42}$$

By exploiting the cyclic property of the trace operator,

$$X^T PB\tilde{W}^T h = \operatorname{tr}\left(\tilde{W}^T hX^T PB\right), \tag{43}$$

the derivative of the Lyapunov function can be rewritten as

$$\begin{aligned}\dot{V} =& -\frac{1}{2}X^T QX + X^T PB\varepsilon \\ &+ \frac{1}{\gamma}\operatorname{tr}\left[\tilde{W}^T\left(\gamma hX^T PB - \dot{\hat{W}}\right)\right].\end{aligned} \tag{44}$$

B. Adaptive Law with σ-Modification

To eliminate the neural-network parameter estimation term in (42), the following trace identity is first employed:

$$X^T PB\tilde{W}^T h(x) = \operatorname{tr}\left[\tilde{W}^T h(x)X^T PB\right]. \tag{45}$$

Substituting (45) into (42) yields

$$\dot{V} = -\frac{1}{2}X^T QX + X^T PB\varepsilon + \frac{1}{\gamma}\operatorname{tr}\left[\tilde{W}^T\left(\gamma h(X)X^T PB - \dot{\hat{W}}\right)\right]. \tag{46}$$

Accordingly, the adaptive law is selected as

$$\dot{\hat{W}} = \gamma h(X)X^T PB, \tag{47}$$

which completely cancels the parameter estimation term in the Lyapunov derivative. Consequently,

$$\dot{V} = -\frac{1}{2}X^T QX + X^T PB\varepsilon. \tag{48}$$

Since the neural-network approximation error is bounded, namely,

$$\|\varepsilon\| \le \varepsilon_0, \tag{49}$$

the Lyapunov derivative satisfies

$$\dot{V} \le -\frac{1}{2}\lambda_{\min}(Q)\|X\|^2 + \|PB\|\varepsilon_0\|X\|. \tag{50}$$

Therefore, whenever

$$\|X\| \ge \frac{2\|PB\|\varepsilon_0}{\lambda_{\min}(Q)}, \tag{51}$$

one has $\dot{V} \le 0$. Hence, the closed-loop system is uniformly ultimately bounded.

Remark 3: The standard adaptive law may suffer from the parameter drift problem when persistent disturbances or approximation errors exist. Since the tracking error only converges to a bounded neighborhood of the origin, the weight update may continue accumulating residual errors, resulting in unbounded parameter estimation.

To further improve the robustness of the adaptive law and suppress parameter drift caused by approximation errors and external disturbances, the following σ-modification is introduced

$$\dot{\hat{W}} = \gamma h(X)X^T PB - \sigma\gamma\|X\|\hat{W}, \tag{52}$$

where $\sigma > 0$ denotes the leakage coefficient.

Substituting (52) into (46) gives

$$\dot{V} = -\frac{1}{2}X^T QX + X^T PB\varepsilon + \sigma\|X\|\operatorname{tr}\left(\tilde{W}^T\hat{W}\right). \tag{53}$$

C. Uniform Ultimate Boundedness Analysis

To analyze the influence of the σ-modification term, the following identity is first established:

$$\operatorname{tr}\left(\tilde{W}^T\hat{W}\right) = \operatorname{tr}\left(\tilde{W}^T W^*\right) - \operatorname{tr}\left(\tilde{W}^T\tilde{W}\right). \tag{54}$$

Applying the Cauchy–Schwarz inequality gives

$$\operatorname{tr}\left(\tilde{W}^T W^*\right) \le \|\tilde{W}\|_F \|W^*\|_F \le \|\tilde{W}\|_F W_{\max}, \tag{55}$$

where $W_{\max}$ is a positive constant satisfying

$$\|W^*\|_F \le W_{\max}.$$

Substituting (54) and (55) into (53) yields

$$\begin{aligned}\dot{V} \le &-\frac{1}{2}\lambda_{\min}(Q)\|X\|^2 + \|PB\|\varepsilon_0\|X\| \\ &+ \sigma\|X\|\|\tilde{W}\|_F W_{\max} - \sigma\|X\|\|\tilde{W}\|_F^2.\end{aligned} \tag{56}$$

Factoring out $\|X\|$ gives

$$\begin{aligned}\dot{V} \le -\|X\|\Big[&\frac{1}{2}\lambda_{\min}(Q)\|X\| + \sigma\|\tilde{W}\|_F^2 \\ &- \sigma W_{\max}\|\tilde{W}\|_F - \|PB\|\varepsilon_0\Big].\end{aligned} \tag{57}$$

Completing the square gives

$$\begin{aligned}\sigma\|\tilde{W}\|_F^2 - \sigma W_{\max}\|\tilde{W}\|_F =& \sigma\left(\|\tilde{W}\|_F - \frac{1}{2}W_{\max}\right)^2 \\ &- \frac{\sigma}{4}W_{\max}^2.\end{aligned} \tag{58}$$

Substituting (58) into (57) yields

$$\begin{aligned}\dot{V} \le -\|X\|\Bigg[&\frac{1}{2}\lambda_{\min}(Q)\|X\| + \sigma\left(\|\tilde{W}\|_F - \frac{1}{2}W_{\max}\right)^2 \\ &- \left(\frac{\sigma}{4}W_{\max}^2 + \|PB\|\varepsilon_0\right)\Bigg].\end{aligned} \tag{59}$$

Therefore,

$$\|X\| > \frac{2}{\lambda_{\min}(Q)}\left(\frac{\sigma}{4}W_{\max}^2 + \|PB\|\varepsilon_0\right), \tag{60}$$

and

$$\|\tilde{W}\|_F > \frac{1}{2}W_{\max} + \sqrt{\frac{1}{4}W_{\max}^2 + \frac{\|PB\|\varepsilon_0}{\sigma}}, \tag{61}$$

imply that

$$\dot{V} < 0.$$

Hence, both the system state X and the neural-network weight estimation error $\tilde{W}$ are uniformly ultimately bounded.

Remark 4: The conditions in (60) and (61) are alternative conditions. The satisfaction of either condition is sufficient to guarantee that $\dot{V} < 0$, which ensures the uniform ultimate boundedness of the closed-loop system.

Consequently, all closed-loop signals remain bounded, and the tracking error converges to a compact neighborhood of the origin despite the presence of modeling uncertainties and external disturbances.

TABLE II: Definitions of Control Modes and Adaptive Laws

Parameter	Value	Description
S_1	1	First adaptive law
S_1	2	Second adaptive law with σ-modification
S	1	Nominal model-based controller without compensation
S	2	Controller with exact uncertainty compensation
S	3	Controller with RBF neural network compensation

VI. Simulation

Numerical simulations are conducted to verify the effectiveness of the proposed RBF neural network-based adaptive control strategy. Two nonlinear systems with different uncertainty characteristics are considered. The first example investigates the approximation capability of the RBF compensator under non-smooth disturbances, while the second example evaluates the proposed controller for a multi-joint robotic manipulator with dynamic coupling and lumped uncertainties.

A. Example 1: Nonlinear Servo System with Non-Smooth Disturbance

Consider a single-degree-of-freedom nonlinear servo system described by

$$M\ddot{q} = \tau + d(\dot{q}), \tag{62}$$

where $M = 10$ and the disturbance term is given by

$$d(\dot{q}) = -15\dot{q} - 30\,\operatorname{sgn}(\dot{q}). \tag{63}$$

Before presenting the simulation results, the overall simulation configuration is introduced. The Simulink implementation of the proposed control framework is illustrated in Fig. 1. The simulation platform consists of the nonlinear servo system, the controller module, the reference trajectory generator, and the uncertainty compensation module.

The controller receives the desired trajectory and the measured system states, and generates the control input τ. Depending on the selected control mode, the uncertainty compensation term is either disabled ($S = 1$), obtained from the known disturbance model ($S = 2$), or generated by the RBF neural network ($S = 3$).

The disturbance consists of viscous friction and Coulomb friction components, which introduces a non-smooth uncertainty into the closed-loop system.

The desired trajectory is selected as

$$q_d(t) = \sin(t)\ \text{rad}, \tag{64}$$

with the initial condition

$$[q(0), \dot{q}(0)]^T = [0.6, 0]^T.$$

The controller parameters are selected as

$$Q = 50I_2, \qquad \gamma = 200, \qquad \sigma = 0.001.$$

The RBF neural network employs Gaussian basis functions with different numbers of hidden nodes. Two configurations with $m = 5$ and $m = 19$ are considered.

For the case of $m = 5$, the centers are selected as

$$c_i = [\alpha_i, \alpha_i]^T,$$

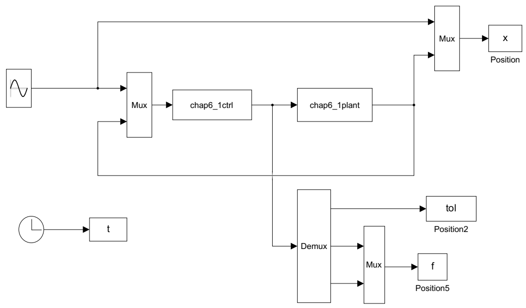

Fig. 1: Simulink implementation of the nonlinear servo system and the proposed control framework.

where

$$\alpha_i \in \{-1, -0.5, 0, 0.5, 1\}.$$

For the case of $m = 19$, a denser distribution of centers is adopted:

$$\alpha_i \in \{-1, -8/9, \ldots, 0, \ldots, 8/9, 1\}.$$

The Gaussian basis functions are defined as

$$h_i(\xi) = \exp\left(-\frac{\|\xi - c_i\|^2}{2b^2}\right),$$

where the width parameter is selected as $b = 5$ for all basis functions. The initial neural network weights are set as $\hat{W}(0) = 0$.

To evaluate the effectiveness of the proposed adaptive compensation strategy, three different compensation modes are considered:

- $S = 1$: nominal model-based controller without uncertainty compensation;
- $S = 2$: exact compensation controller assuming that the uncertainty $d(\dot{q})$ is completely known;
- $S = 3$: RBF neural network compensation controller, where the unknown uncertainty is approximated online.

For $S = 3$, two adaptive laws are investigated: $S_1 = 1$ and $S_1 = 2$.

1) Nominal Model-Based Controller ($S = 1$): In this case, only the nominal model is used in the controller design. The disturbance term is not compensated, and the adaptive law selection parameter S_1 has no influence because the RBF network is not involved.

The tracking response is shown in Fig. 2, while the uncertainty response is presented in Fig. 3.

As observed in Fig. 2, the nominal controller cannot completely reject the non-smooth friction disturbance, resulting in a non-zero tracking error.

Since no compensation mechanism is activated, the estimated uncertainty remains zero, as shown in Fig. 3.

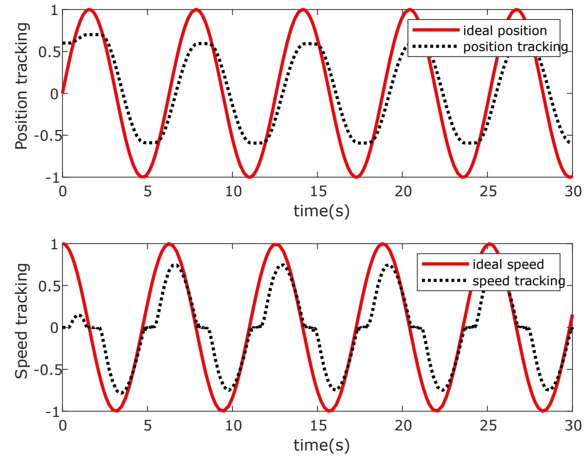

Fig. 2: Tracking performance of the nominal controller ($S = 1$).

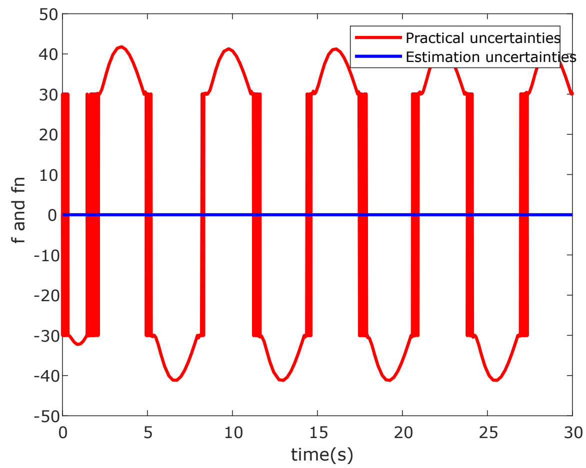

Fig. 3: Unknown disturbance and estimation result under the nominal controller ($S = 1$).

2) Exact Compensation Controller ($S = 2$): For $S = 2$, the uncertainty term is assumed to be known exactly. Therefore, the compensation input directly cancels the disturbance:

$$\tau_c = -d(\dot{q}).$$

The adaptive law parameter S_1 does not affect the control performance because the RBF neural network is not used.

As shown in Fig. 4, the tracking error converges rapidly because the disturbance is completely compensated.

Different from the RBF compensation case, the uncertainty estimation is not generated by the neural network. Instead, the exact disturbance model is directly used for compensation.

3) RBF Neural Network Compensation ($S = 3$): When $S = 3$, the explicit expression of the nonlinear disturbance is assumed to be unknown to the controller. Therefore, an RBF neural network is introduced to approximate the lumped uncertainty:

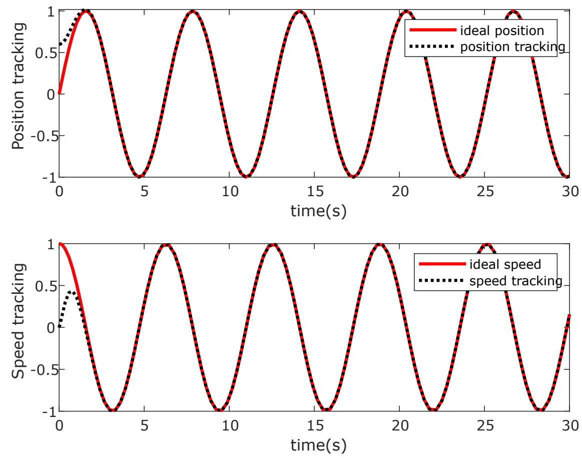

Fig. 4: Tracking performance with exact uncertainty compensation ($S = 2$).

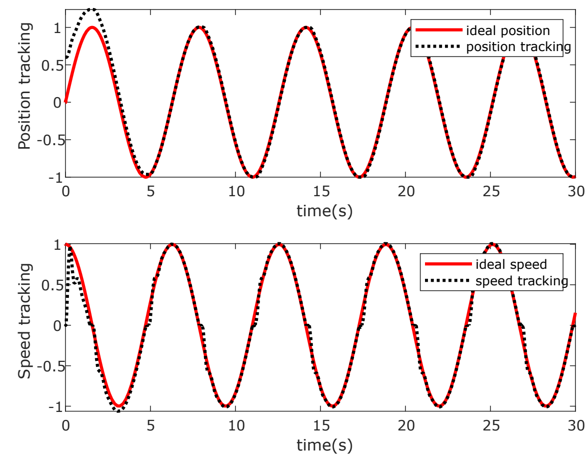

Fig. 6: Tracking performance of the RBF adaptive controller with the first adaptive law ($S = 3, S_1 = 1$).

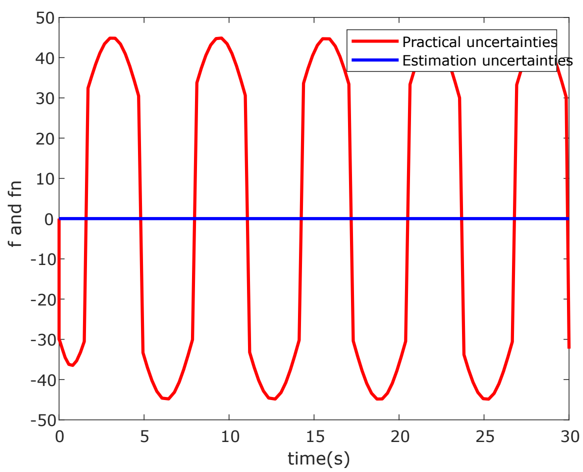

Fig. 5: Exact disturbance compensation result ($S = 2$).

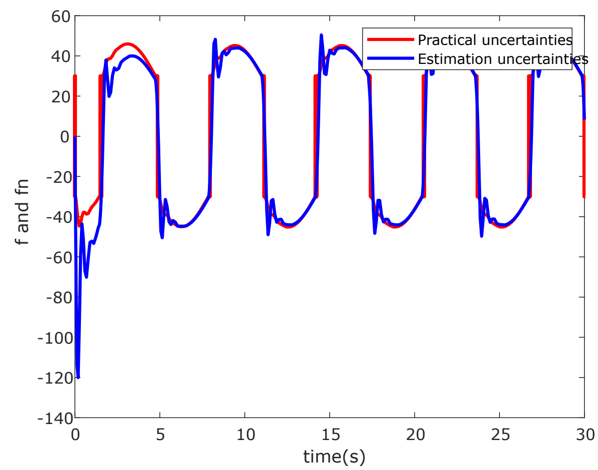

Fig. 7: Approximation of the nonlinear disturbance using the RBF neural network with the first adaptive law ($S = 3, S_1 = 1$).

$$\hat{f}(x) = \hat{W}^T h(\xi), \tag{65}$$

where $\hat{W}$ denotes the adaptive weight vector and $h(\xi)$ represents the Gaussian basis function vector.

Unlike the previous two cases, the disturbance compensation is generated by the neural network rather than directly using the known disturbance model. Therefore, the adaptive law plays an important role in determining the convergence behavior of the estimated uncertainty.

a) First adaptive law ($S = 3, S_1 = 1$): Firstly, the RBF controller with the first adaptive law is investigated. The tracking performance and uncertainty approximation results are shown in Fig. 6 and Fig. 7, respectively.

As shown in Fig. 6, the RBF compensation effectively reduces the tracking error compared with the nominal controller. The approximation result in Fig. 7 demonstrates that the neural network is able to learn the unknown disturbance online.

b) Second adaptive law with $m = 5$ hidden nodes ($S = 3, S_1 = 2$): Next, the second adaptive law is evaluated with an RBF neural network containing five hidden nodes.

The corresponding tracking response and disturbance approximation result are shown in Fig. 8 and Fig. 9.

With a limited number of hidden nodes, the RBF network provides a coarse approximation of the nonlinear disturbance. The residual approximation error leads to a relatively larger tracking deviation.

c) Second adaptive law with $m = 19$ hidden nodes ($S = 3, S_1 = 2$): To further investigate the influence of the approximation capability, the number of hidden nodes is increased to $m = 19$.

The corresponding results are presented in Fig. 10 and Fig. 11.

Compared with the case of $m = 5$, the denser distribution of Gaussian basis functions improves the approximation capability of the RBF neural network. Consequently, the estimated uncertainty is closer to the actual disturbance, which leads to improved tracking performance.

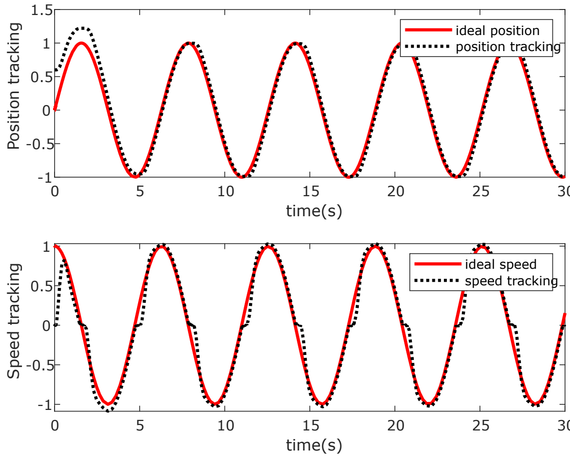

Fig. 8: Tracking response using the second adaptive law with five hidden nodes ($S = 3, S_1 = 2, m = 5$).

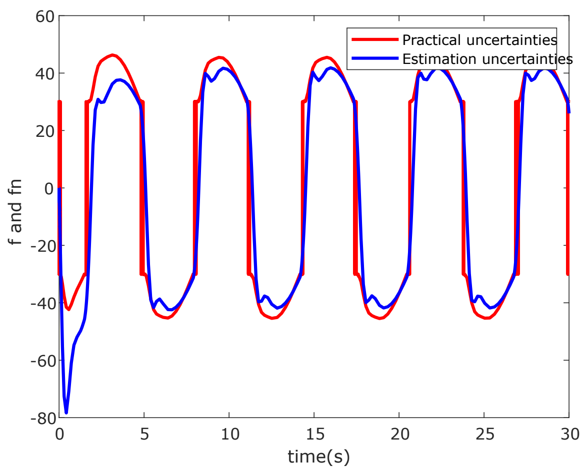

Fig. 9: RBF approximation result with five hidden nodes ($S = 3, S_1 = 2, m = 5$).

d) Summary: The simulation results verify the effectiveness of the RBF-based adaptive compensation strategy.

When the uncertainty model is unavailable, the RBF neural network can successfully approximate the unknown nonlinear disturbance and reduce the tracking error.

Furthermore, increasing the hidden-layer size from $m = 5$ to $m = 19$ enhances the approximation capability of the neural network. The comparison also shows that the adaptive law selection affects the weight update behavior and uncertainty estimation performance.

4) Discussion: The simulation results verify the effectiveness of different compensation strategies.

The nominal controller ($S = 1$) suffers from residual tracking errors because the disturbance is completely uncompensated.

The exact compensation controller ($S = 2$) achieves the best tracking performance since the disturbance is assumed to be completely known.

For the practical case where the uncertainty model is

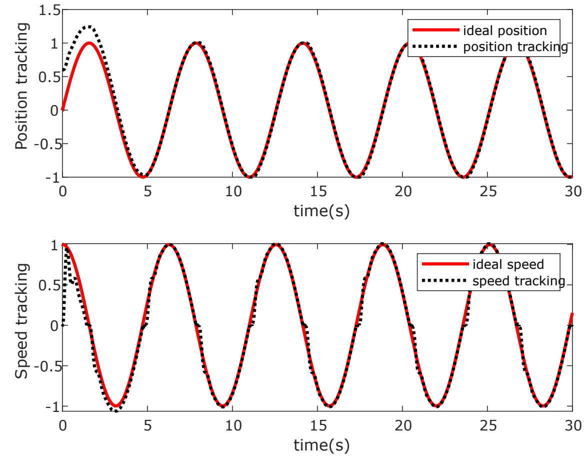

Fig. 10: Tracking response using the second adaptive law with nineteen hidden nodes ($S = 3, S_1 = 2, m = 19$).

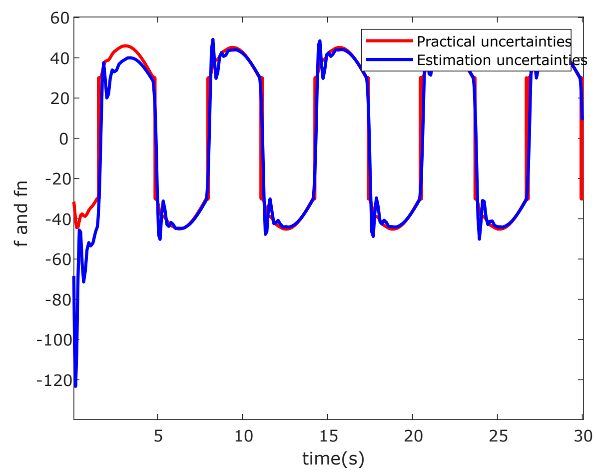

Fig. 11: RBF approximation result with nineteen hidden nodes ($S = 3, S_1 = 2, m = 19$).

unavailable, the RBF-based adaptive controller ($S = 3$) successfully approximates the unknown disturbance and achieves improved tracking performance.

Moreover, the comparison between $S_1 = 1$ and $S_1 = 2$ demonstrates that the second adaptive law provides improved weight regulation while preserving tracking accuracy.

B. Example 2: Two-Link Robotic Manipulator with Coupled Uncertainties

A two-link planar robotic manipulator is considered to further evaluate the proposed adaptive control strategy in the presence of nonlinear dynamic coupling, parameter uncertainties, and external disturbances. The simulation configuration implemented in Simulink is shown in Fig. 12.

The nominal dynamics of the robotic manipulator are described as

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau + d, \tag{66}$$

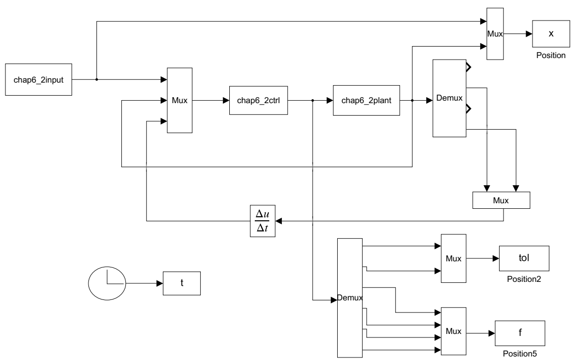

Fig. 12: Simulink implementation of the two-link robotic manipulator control system.

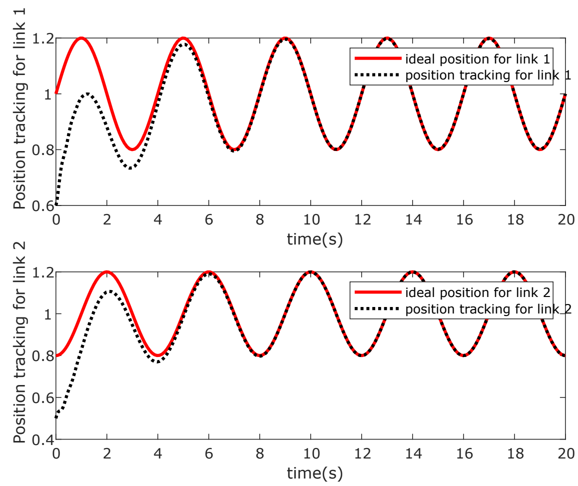

Fig. 13: Angle tracking responses of joint 1 and joint 2 under the proposed RBF adaptive controller ($S = 3, S_1 = 2$).

where the inertia matrix, Coriolis matrix, and gravity vector are given by

$$M(q) = \begin{bmatrix} v + q_{01} + 2q_{02}\cos(q_2) & q_{01} + q_{02}\cos(q_2) \\ q_{01} + q_{02}\cos(q_2) & q_{01} \end{bmatrix}, \tag{67}$$

$$C(q, \dot{q}) = \begin{bmatrix} -q_{02}\dot{q}_2\sin(q_2) & -q_{02}(\dot{q}_1 + \dot{q}_2)\sin(q_2) \\ q_{02}\dot{q}_1\sin(q_2) & 0 \end{bmatrix}, \tag{68}$$

$$G(q) = \begin{bmatrix} 15g\cos(q_1) + 8.75g\cos(q_1 + q_2) \\ 8.75g\cos(q_1 + q_2) \end{bmatrix}. \tag{69}$$

The physical parameters are selected as

$$v = 13.33, \qquad q_{01} = 8.98, \qquad q_{02} = 8.75, \qquad g = 9.8\ \mathrm{m/s^2}.$$

The external disturbance is selected as an error-dependent nonlinear uncertainty:

$$d(t, e, \dot{e}) = d_1 + d_2\|e\| + d_3\|\dot{e}\|, \tag{70}$$

where

$$d_1 = [2, 2]^T, \qquad d_2 = [3, 3]^T, \qquad d_3 = [6, 6]^T.$$

To evaluate the robustness of the proposed controller against modeling uncertainties, a 20% parameter mismatch is introduced into the nominal dynamic model:

$$\Delta M = 0.2M_0, \qquad \Delta C = 0.2C_0, \qquad \Delta G = 0.2G_0.$$

The desired trajectories of the two joints are selected as

$$q_{1d}(t) = 1 + 0.2\sin(0.5\pi t), \tag{71}$$
$$q_{2d}(t) = 1 - 0.2\cos(0.5\pi t). \tag{72}$$

The initial condition is selected as

$$X(0) = [0.6, 0.3, 0.5, 0.5]^T.$$

The control parameters are chosen as

$$Q = 50I_4, \qquad \gamma = 20, \qquad \sigma = 0.001.$$

In this example, the RBF neural network compensation controller with the second adaptive law ($S = 3, S_1 = 2$) is adopted. The Gaussian basis functions are initialized with

$$\hat{W}(0) = 0.1I.$$

The RBF network approximates the unknown lumped uncertainty as

$$\hat{f}(x) = \hat{W}^T h(x).$$

The simulation results are presented in the following figures. The joint angle tracking responses are shown in Fig. 13, while the corresponding angular velocity tracking performance is presented in Fig. 14.

The generated control inputs of the two joints are illustrated in Fig. 15.

Finally, the actual lumped uncertainty and its RBF approximation are shown in Fig. 16.

The simulation results demonstrate that the proposed adaptive RBF compensation controller achieves accurate trajectory tracking despite the existence of nonlinear coupling, parameter uncertainties, and external disturbances.

As shown in Fig. 13 and Fig. 14, both joint positions and angular velocities converge to their desired trajectories with small tracking errors.

Furthermore, Fig. 16 verifies that the RBF neural network can effectively approximate the unknown lumped uncertainty online. The generated control inputs remain bounded and smooth, as illustrated in Fig. 15, demonstrating the robustness of the proposed adaptive compensation strategy.

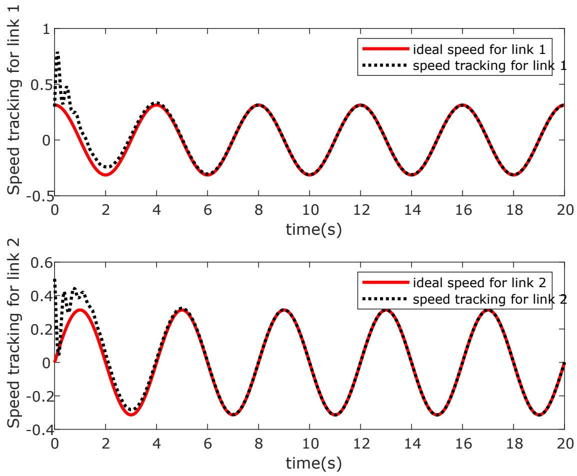

Fig. 14: Angular velocity tracking responses of joint 1 and joint 2.

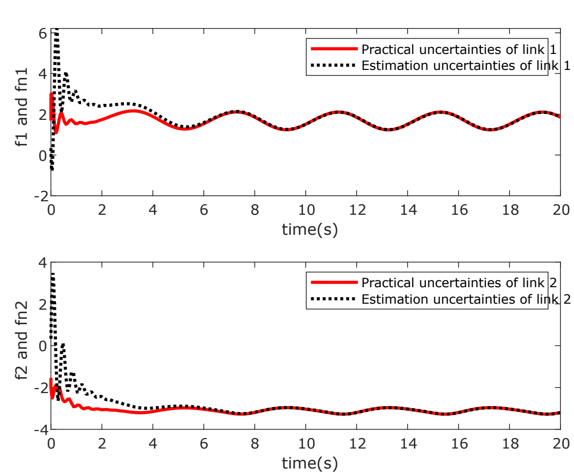

Fig. 16: Actual lumped uncertainty and its RBF neural network approximation.

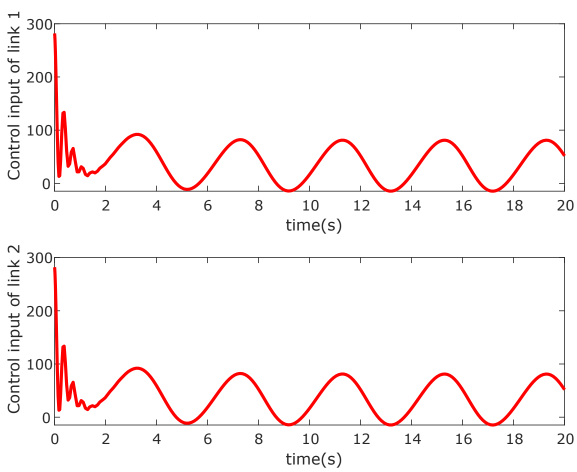

Fig. 15: Control inputs of joint 1 and joint 2 generated by the adaptive controller.

VII. Conclusion

This paper has investigated an adaptive neural network-based control strategy for robotic manipulators subject to modeling uncertainties, external disturbances, and unmodeled nonlinear dynamics. By decomposing the robot dynamics into a nominal model and a lumped uncertainty term, an RBF neural network was introduced to approximate the unknown nonlinear function online. Based on the nominal inverse dynamics and neural compensation, a modular controller structure was developed, where the trajectory tracking objective and uncertainty compensation mechanism are separately designed.

A Lyapunov-based adaptive law was derived to update the neural network weights, and a σ-modification technique was incorporated to improve the robustness against persistent approximation errors and disturbances. The stability analysis demonstrated that the closed-loop system is uniformly ultimately bounded, and all system states and neural network weight estimation errors remain bounded under the proposed adaptive control framework.

Simulation studies were conducted on both a nonlinear servo system and a two-link robotic manipulator. The results of the servo system verified that the RBF neural network can effectively approximate unknown non-smooth disturbances and improve tracking performance compared with the nominal controller without compensation. Furthermore, increasing the number of hidden nodes enhanced the approximation capability of the neural network. The two-link manipulator simulation further demonstrated the effectiveness of the proposed controller in the presence of dynamic coupling, parameter uncertainties, and external disturbances, achieving accurate trajectory tracking while maintaining bounded control inputs.

Future work will focus on experimental validation using a physical robotic platform and further improvements of the adaptive compensation mechanism under more challenging conditions, such as time-varying uncertainties, actuator constraints, and communication limitations in networked robotic systems.